A Note on Geometric Transfer, Dyadic Structuring, and Racinary Transitions in Prime and Semiprime Distributions

This note is intended as a conceptual extension to the main text *Towards a Geometric Elucidation of Squaring the Circle*. It situates the geometric observations developed therein within a broader framework concerning the structuring and distribution of prime numbers and their semiprime products.

The initial investigation arose from an empirical and geometric study of semiprime integers $n = pq$, where the factors p and q are interpreted as angular positions on concentric dyadic circles indexed by powers of two. Within this representation, $n$occupies a well-defined position on its own dyadic circle, while p and q appear as conjugate angular points whose relative disposition exhibits a pendular relationship with respect to n. This configuration reveals stable geometric constraints that persist across scales, suggesting that the arithmetic structure of semiprimes may admit a meaningful geometric encoding.

At first glimpse, it is essential to distinguish between two fundamentally different orders of magnitude within this geometric framework. On one hand, the **algebraic regime**, represented by the 'racinary' operator $\sqrt{2}$, governs the iterative and discrete transfer between structures; it is the logic of the square and the diagonal, which allows for measurable transitions between dyadic levels. On the other hand, the **transcendental regime**, embodied by π and e, represents continuous closure and 'super-generational' growth. The inherent difficulty of squaring the circle stems from the fact that no finite sequence of algebraic (racinary) operations can perfectly resolve a transcendental (circular) value. Thus, the distribution of primes and semiprimes must be understood as an interaction between these two distinct mathematical orders: the algebraic 'bridge' and the transcendental 'curve'.

A fundamental distinction exists between the algebraic "bridge" and the transcendental "curve." While the algebraic represents "what is" (static values), the **transcendental** represents "what is becoming" (the passage to the limit).

- **The Circular Closure π:** This operator enforces continuous closure. The transcendental nature of π explains the impossibility of the quadrature of the circle; no finite sequence of algebraic (racinary) operations can resolve a pure curve.
- **Trigonometric Transition:** The passage from a straight line to a circle almost always "transcends" classical arithmetic. According to the Lindemann-Weierstrass theorem, the sine or cosine of an algebraic value (excluding 0) results in a transcendental value, marking the threshold where arithmetic enters the circular domain.

The main text focuses on the impossibility of reconciling square-based and circle-based geometries, traditionally framed as the quadrature of the circle. In contrast, the present note addresses a complementary question: how geometric operators of transfer arise naturally when discrete dyadic structuring interacts with continuous angular phase. In particular, it highlights the role of racinary scaling—governed by square roots—as an intermediate regime between dyadic powers and circular closure.

Within the interval between two consecutive dyadic circles $2^k$and 2^{k+1}, a structured family of intermediate constructions emerges, not dyadic in nature but governed by powers of $\sqrt{2}$. These constructions correspond to successive circle–square transitions and can be interpreted as geometric transport operators linking discrete arithmetical levels. Unlike the dyadic hierarchy, which defines sharp structural boundaries, this racinary regime encodes the manner in which positions are transferred within a scale, preserving angular information while modifying radial magnitude.

This perspective sheds new light on the behavior of prime numbers and semiprimes: primes appear as points of irreducible asymmetry within the dyadic framework, while semiprimes reveal stable geometric relations between their factors through angular offsets and racinary transport. More broadly, it suggests that certain irrational constants — such as $\sqrt{2}$ and π — should be understood not merely as numerical values, but as generators of distinct geometric regimes: the former enabling iterative transfer between structures, the latter enforcing continuous closure.

The purpose of this note is therefore not to introduce new computational results, but to clarify the conceptual architecture underlying the observed phenomena, and to situate the original geometric construction within a coherent framework linking dyadic scaling, racinary transition, and circular phase. In doing so, it aims to provide a unifying interpretation of the prime and semiprime distributions explored in the main text.

Quest of the Racinary-Concentric Root

The inverse operation corresponding to dyadic multiplication is not a classical root but a racinary-concentric root: an operator that projects the phase $\emptyset n$ back onto admissible conjugate phases (θp, θq) on lower-level circles, respecting carry boundaries, regime transitions, and phase-transport compatibility. It is multi-branched, regime-conditioned, and mediated by square-root-like scalings that bridge dyadic strata while attempting to preserve angular information.

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